



CATHOLIC JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATIONS
Higher 2

**CANDIDATE
NAME**

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CLASS

2T

**INDEX
NUMBER**

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BIOLOGY

Paper 3

9744/03
12 SEPT 2022
2 hours

Candidates answer on the Question Paper.
Additional Materials: Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs. Do not use staples, paper clips, glue or correction fluid.

Section A

Answer **all** questions in the spaces provided on the Question paper.

Section B

Answer **one** question in this section on writing papers provided.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten your answers for **Section B** securely.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1 [25]	
2 [12]	
3 [13]	
4 or 5 [25]	
TOTAL	75

Section A

Answer **all** questions in this section.

Answer **all** questions.

- 1 Climate change influences how mosquito-borne diseases such as dengue, chikungunya and zika infection have spread across the world vectors such as mosquitoes. These mosquitoes generally have a narrow temperature tolerance and there is evidence to show the spread of such diseases migrating from tropical to sub-tropical regions due to global warming.

The effects of temperature on the rate of reaction are expressed as the temperature coefficient (Q_{10}). Changes in temperatures affect enzyme activity which will affect insect's growth and development.

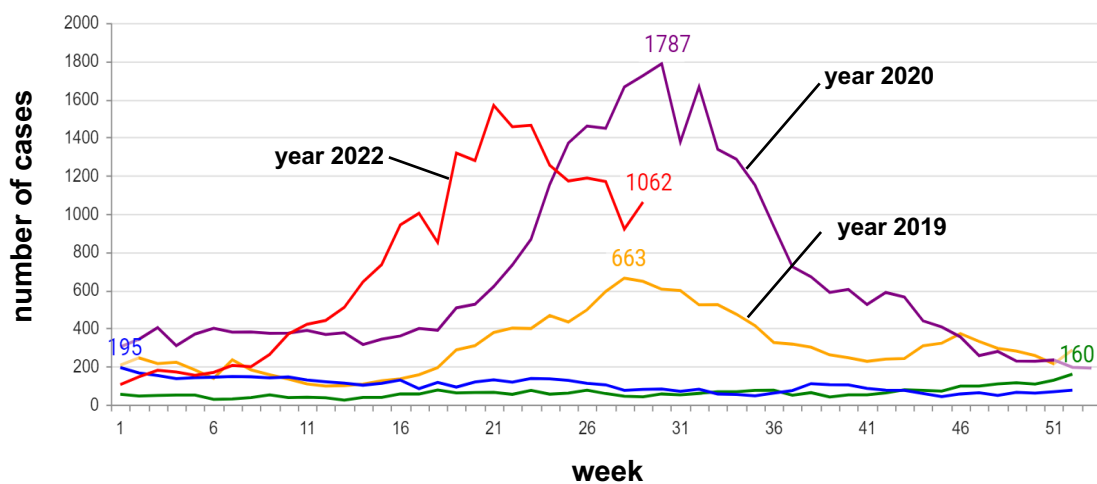
- (a) Define what is Q_{10} theory and explain why insects have a narrow temperature tolerance range as compared to mammals.

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[2]

Singapore's dengue cases could rise even higher than the current number as the nation has entered the traditional peak dengue season, which usually lasts from June to October, corresponding to hotter and wetter climate.

Fig 1.1 shows the weekly number of dengue cases in Singapore from 2018 to 2022.



(Source: NEA website, July 22 - <https://www.nea.gov.sg/dengue-zika/dengue/dengue-cases>
 Communicable Diseases Division, Ministry of Health)

Fig. 1.1

- (b) Explain how Singapore's hotter and wetter climate affects mosquitoes.

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Singapore has been looking on ways to reduce the numbers of *A. aegypti*. Besides project *Wolbachia*, Singapore may consider another type of genetically modified (GM) male mosquitoes. One of the genes added to these mosquitoes, when switched on, results in the production of a protein which is toxic to mosquitoes.

- (c) GM male mosquitoes were released into the wild to mate with females and all the resulting offspring died in the larval stage.

- (i) Suggest why releasing such large numbers of male mosquitoes did not immediately increase the risk of transmission of dengue fever.

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- (ii) Both GM male and non-GM male mosquitoes were released to compare their dispersal and life span in the wild. The GM mosquitoes could be identified because they also carried a gene for green fluorescent protein (GFP).

Suggest why, in many examples of gene technology, fluorescent markers are used.

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In addition to the gene for GFP, the DNA that has been added to the GM mosquitoes consists of

- a promoter
- a gene coding for a toxic protein, tTA
- a binding site for tTA.

When a GM mosquito larva hatches from an egg, the promoter induces the production of only a small amount of tTA, so that the larva does not die immediately. In a process of positive feedback, the tTA produced binds to the DNA as shown in Fig. 1.4. This increases the expression of the gene until the increased concentration of tTA kills the larva.

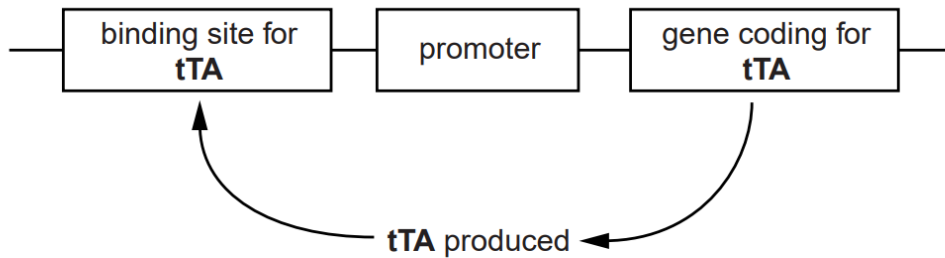


Fig. 1.4

- (d) Explain why this process is called positive feedback and why all 3 components shown in Fig. 1.4 are necessary for positive feedback to occur.

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Dengue fever is a disease caused by infection with a virus transmitted by the *A. aegypti* mosquito. It is an acute viral infection characterised by fever, rash, headache, and muscle and joint pain. Occasionally, dengue virus infections progress to dengue haemorrhagic fever (DHF) and dengue shock syndrome (DSS). Fig 1.5 shows the structure of a dengue virus. Upon entry to host's cells, viral replication and translation begins.

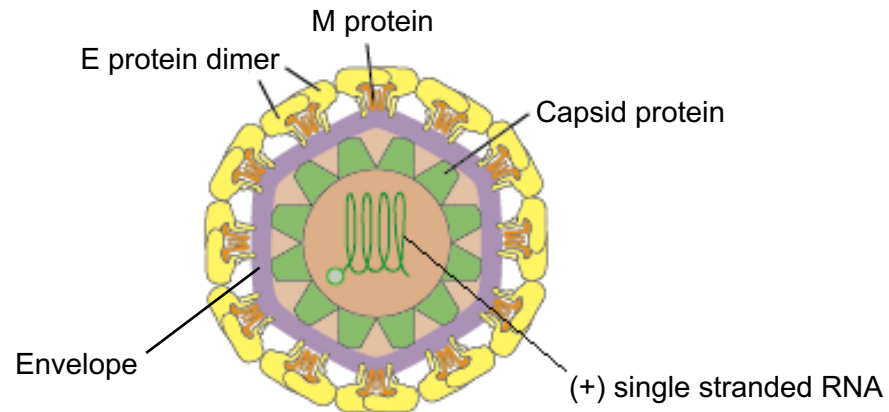


Fig. 1.5

(e) With reference to Fig. 1.5 and your knowledge of influenza virus,

(i) state two ways in which the structure of dengue virus is similar to that of influenza virus.

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(ii) outline how dengue virus replicates within the host's cells to form more viral genome and proteins.

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Dengue viruses consist of four serotypes, DENV-1 to DENV-4. The rapid identification of dengue virus serotypes isolated from patients' blood is important for clinical investigations. Individuals who become infected with one serotype obtain lifelong immunity against that serotype but not the other three.

Fig. 1.6 shows an IgM Antibody Capture Enzyme-Linked Immunosorbent Assay (MAC-ELISA), a molecular tool used to test on dengue patients, who developed anti-DENV antibodies. The tool includes:

- 1) the microtiter plate with anti-human-IgM antibody,
- 2) DENV antigens,
- 3) anti-DENV antibody conjugated with enzyme,
- 4) substrate which can change colour.

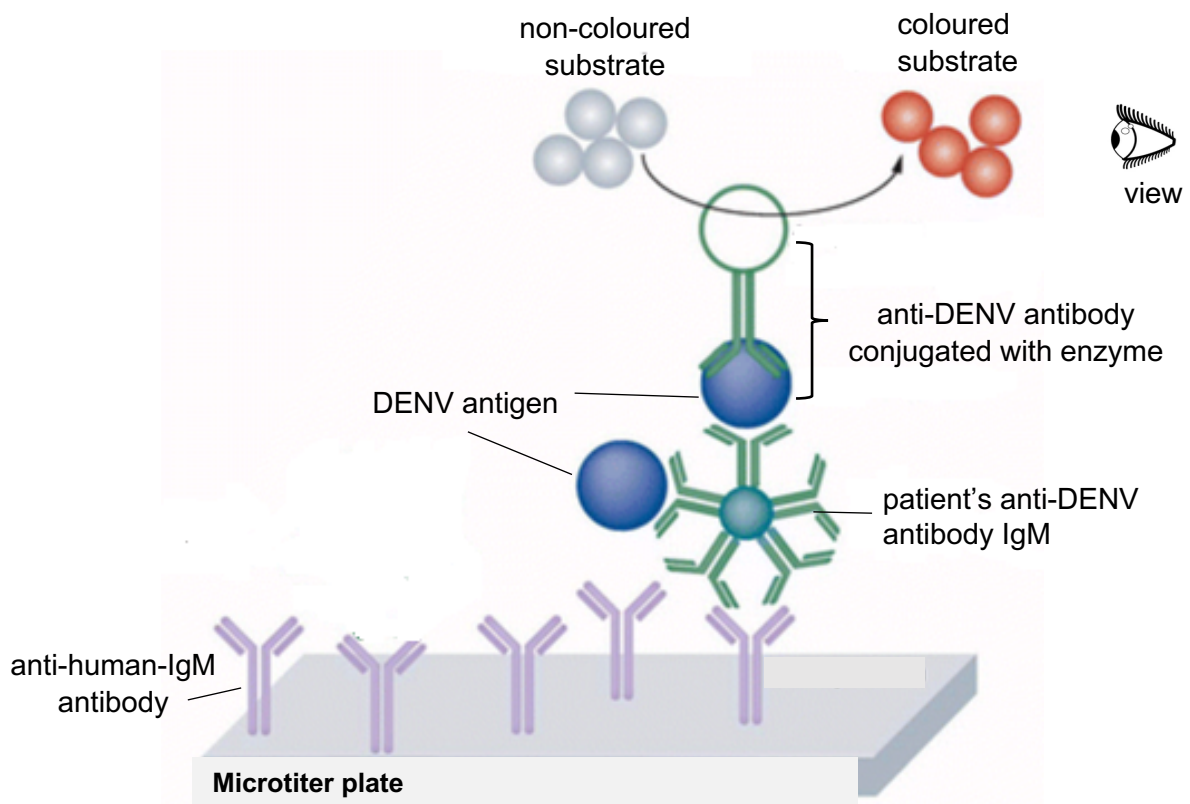


Fig. 1.6

- (f) With reference to Fig. 1.6, suggest the principles of using MAC-ELISA to detect a positive result of a patient infected by the dengue virus

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- (g) Dengue fever is considered an infectious disease, different from other non-infectious diseases like genetic diseases.

Comment on the differences between an infectious disease such as dengue fever to that of a genetic disease, using a named example.

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Scientists have analysed the respiration rates on crickets, *Hophlosphyrum griseus*, another type of insect, subjected to three different ambient temperatures. Fig. 1.7 shows the metabolic rate of crickets was significantly correlated to their body mass.

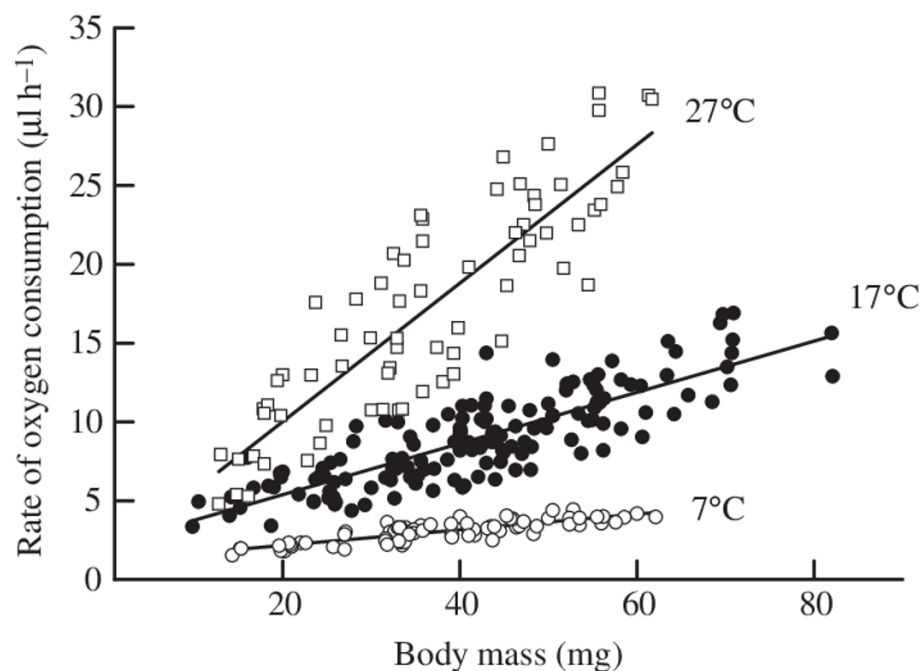


Fig. 1.7

(h) With reference to Fig. 1.7,

- (i) describe the extent to which a cricket's body mass can predict their rate of oxygen consumption at different ambient temperatures.

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- (ii) Outline the role of oxygen in oxygen consumption of these crickets.

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[Total: 25]

2 Stem cells have an important role in maintenance and growth within the bodies of eukaryotic organisms.

(a) Explain how stem cells maintain their presence in tissues.

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(b) Normal differentiated cells usually have a limit to the number of cell divisions that can take place before apoptosis. Explain why stem cells are different and are able to divide indefinitely.

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Embryonic stem cells, which are derived from the inner cell mass of mammalian blastocysts, are used in research to understand disease development and develop future cell-based therapies for currently untreatable diseases.

In 2006, a team of scientists led by Shinya Yamanaka discovered that differentiated cells can be isolated and modified to an embryonic-like state. These cells, subsequently designated as induced pluripotent stem cells (iPSCs), have been hailed as an effective replacement for human embryonic stem cells for its usefulness in regenerative medicine. Fig. 2.1 shows a potential application of human iPSCs in gene therapy.

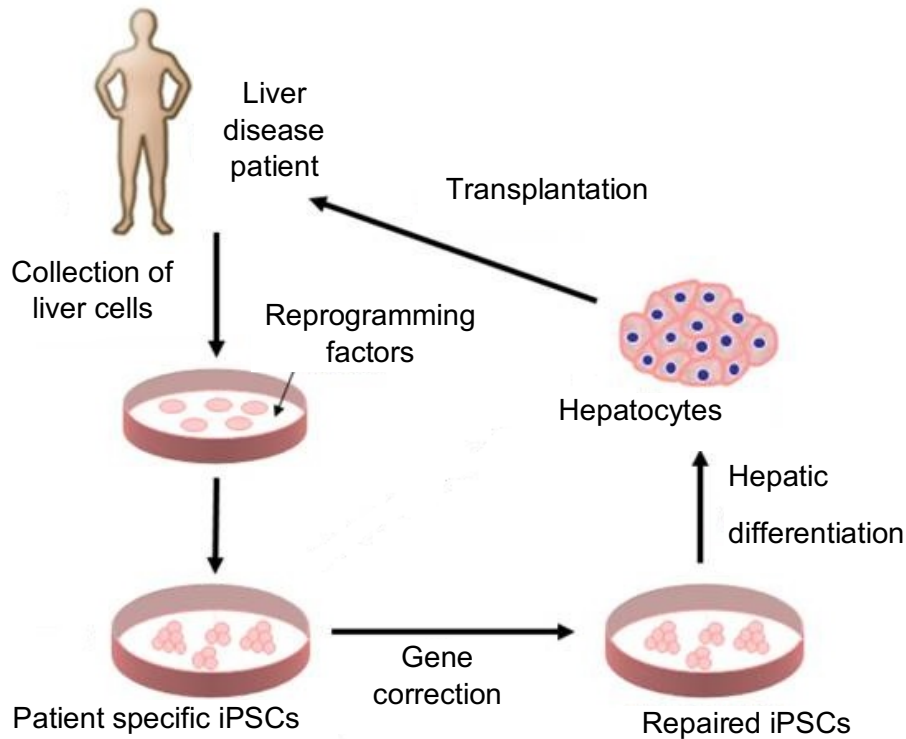


Fig. 2.1

- (c) Embryonic stem cells and iPSC are both pluripotent. In terms of genetic control, suggest how this is possible.

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- (d) With reference to Fig. 2.1 state and explain two advantages of using iPSCs in gene therapy.

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- (e) Yamanaka's research team studied 24 genes expressed by embryonic stem cells in mice to track down genes that can induce pluripotency. They discovered that four genes, notably *Oct4*, *Sox2*, *Klf4* and *c-Myc*, encode transcription factors which could be used to reprogram differentiated cells to form iPSCs. These genes can be introduced into differentiated somatic cells via viral transduction or the introduction of non-integrating plasmids in Fig. 2.2.

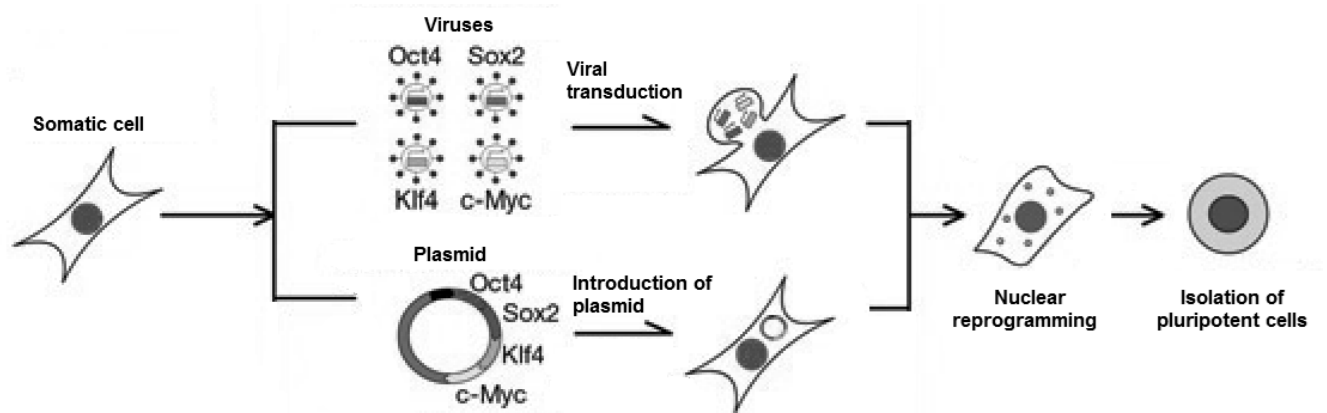


Fig 2.2

- (i) Suggest why it is necessary to introduce these four genes into somatic cells in iPSC formation.

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- (ii) Describe one advantage and one disadvantage of introducing the four genes via non-integrating plasmids over viral transduction using retroviruses.

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[Total: 12]

- 3** Viruses are often considered to be in the grey area of life. Over the last 15 years or so, giant viruses found in amoebae have complicated their status as simple non-living structures. Mimiviruses and megaviruses can contain more genes than a simple bacterium and may encode genes for information storage and processing. Genes common to the domains Archaea, Bacteria and Eukarya can be found in different giant viruses, and some researchers argue on this basis that they constitute a fourth domain of life.

This view challenges the cell theory, where the three tenets are as described below:

1. All living organisms comprise one or more cells.
2. The cell is the smallest unit of life.
3. All cells arise from pre-existing cells.

- (a)** Discuss how viruses challenge the cell theory and concepts of what is considered living.

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- (b)** Despite being considered as non-living things, viruses are known to mutate and evolve into different strains. Concerns over the effectiveness of vaccines against the inevitable evolution of the coronavirus broke out amidst the pandemic.

In Singapore, one of common type of vaccine used for the coronavirus is also known as the mRNA vaccine.

1. First, mRNA COVID-19 vaccines are given in the upper arm muscle or upper thigh, depending on the age of who is getting vaccinated.
2. After vaccination, the mRNA will enter the muscle cells.
3. Once inside, they use the cells' machinery to produce a harmless piece of what is called the spike protein. The spike protein is found on the surface of the virus that causes COVID-19.

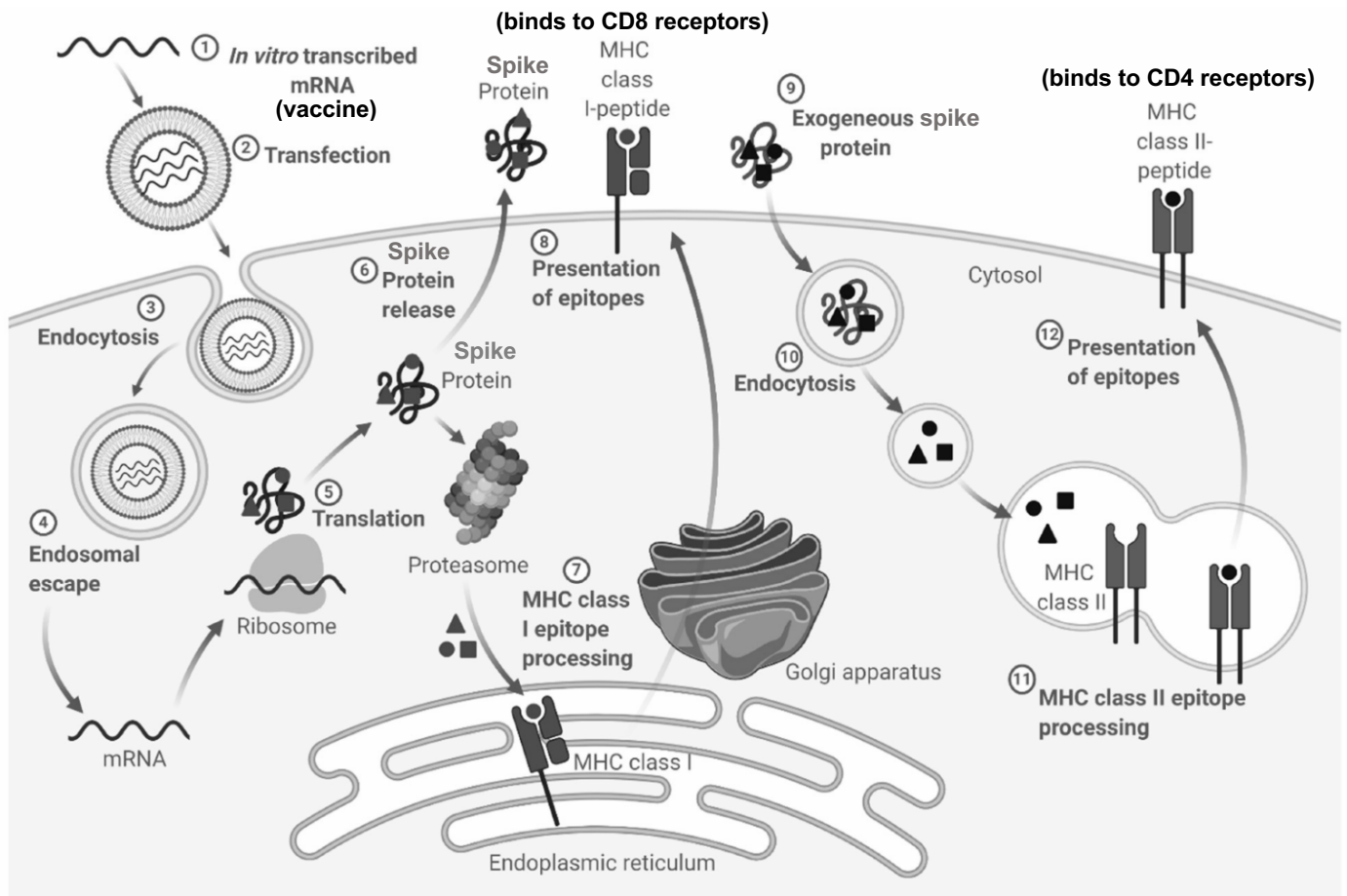


Fig. 3.1

- (i) With reference to the information above and Fig. 3.1, outline how an mRNA vaccine allows an acquired immune memory to be built against future coronavirus infection.

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- (ii) Explain why, despite not being considered as living organisms, viruses can evolve.

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Concerns over the efficiency of vaccines was not just because of the potential ineffectiveness against new viral strains, but the consequences of expanding partial dosage of the vaccines in an attempt to increase the number of people able to receive the vaccine.

Concerns were raised about having a fraction of the population with partial immunity to the virus could potentially increase selection for vaccine-escape variants, ultimately undermining vaccine effectiveness.

- (iii) Discuss if there should be concerns about the effectiveness of vaccines, given that the coronavirus can evolve.

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[Total: 13]

Section B

Answer **one** question in this section.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

- 4 (a) Describe the molecular structure and functions of enzymes, and explain why high fevers can be dangerous for the body. [13]
- (b) Explain the significance of membranes in ATP synthesis in living organisms, and discuss the significance of ATP synthesis in organisms that lack such membranes. [12]
- 5 (a) Outline how eukaryotic somatic cells normally divide, and describe how this process could develop into one that results in cancerous cells instead. [14]
- (b) Despite selection pressures selecting for certain phenotypes, some phenotypes continue to persist in a population.
- Explain how both genetic and environmental factors contribute to phenotypic expressions in eukaryotes and suggest how phenotypes which are **selected against** could be preserved in a population. [11]

[End of Paper 3]

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